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Purine Nucleoside Phosphorylase from Human Erythrocytes

I. PURIFICATION AND PROPERTIES*

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SUMMARY

Purine nucleoside phosphorylase (purine nucleoside:orthophosphate ribosyltransferase, EC 2.4.2.1) from human erythrocytes has been purified about 7000-fold to a specific activity of 80 μM units per mg of protein. At this state of purity, single coincident peaks of enzymatic activity and protein emerged from Sephadex G-100 columns. Use of the molecular sieve method of Andrews permitted estimation of the molecular weight at about 80,000. The enzyme is inactivated by thiol reagents such as Ellman's reagent or *p*-chlormercuribenzoate, and the lost activity may be partially restored by thiols such as mercaptoethanol or dithiothreitol. In addition to its phosphorolytic activity, the enzyme catalyzes a slow ribosyl transfer reaction in the absence of inorganic orthophosphate. The ratio of these activities remained relatively constant throughout purification, and it was not possible to detect any separation of these activities. Kinetic measurements with inosine and deoxyinosine as substrates revealed linear Lineweaver-Burk plots at low concentration, but downward curvatures as the concentration was increased. The results of alternative substrate studies were consistent with the hypothesis that the enzyme is multivalent with cooperative interaction between the active sites.

The enzymatic degradation of purine nucleosides has been known since 1911 (1). In 1945, Kalckar made the important discovery that ribose 1-phosphate is one of the products of the purine nucleoside phosphorylase reaction (2),

$$\text{Purine ribonucleoside} + P_i \rightleftharpoons \text{purine base} + \text{ribose 1-phosphate} \quad (1)$$

Many studies of this and related enzymes have been performed, and the earlier investigations have been thoroughly reviewed by Friedkin and Kalckar (3). Purine nucleoside phosphorylase (purine nucleoside:orthophosphate ribosyltransferase, EC 2.4.2.1) has been identified and studied in a number of mammalian tissues, and the enzymes investigated to date have certain properties in common. The natural purine bases, guanine, hypoxanthine, and xanthine, as well as certain related structural analogues, such as 8-azaguanine, serve as substrates. However, adenine is inactive as a substrate. The sugar moiety may be either ribose or 2-deoxyribose; in the latter case, the product formed is deoxyribose 1-phosphate.

In addition to the above reaction, purine nucleoside phosphorylases from mammalian sources have been reported to catalyze purine base exchange or ribosyl transfer reactions in the absence of inorganic orthophosphate (4), and exchange between orthophosphate and ribose 1-phosphate in the absence of purine base (3, 5). Also, it has recently been reported that, with purine nucleoside phosphorylase from Ehrlich ascites cells, an interconversion between phosphorylase and ribosyl transfer activities can occur, accompanied by marked changes in the molecular weight of the enzyme (6). These observations have raised many interesting questions concerning the reaction mechanism, and suggest that a ribosylated or phosphorylated enzyme might serve as an intermediate, although evidence has been presented in the past which opposes these concepts (7).

Purine nucleoside phosphorylase occurs in high concentration in human erythrocytes, where it appears to play an important role in the metabolism of these cells. Significant advances have been made in the isolation of purine nucleoside phosphorylase from this tissue (8, 9), and therefore human red blood cells were chosen as a source of the enzyme. A preliminary report of this work has been presented (10).

EXPERIMENTAL PROCEDURE

Materials

Inosine, guanine, hypoxanthine, dicyclohexylammonium ribose 1-phosphate, and crystalline purine nucleoside phosphorylase from calf spleen (specific activity, 40 enzyme units per mg) were purchased from Boehringer Mannheim. Crystalline bovine albumin was a product of Armour Pharmaceutical Company. Xanthine oxidase was obtained from Worthington. DEAE-

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cellulose (Cellex-D), dithiothreitol, dGDP, deoxyinosine, deoxyguanosine, and guanosine were purchased from Calbiochem. Sephadex was obtained from Pharmacia. Whatman chromatography cellulose powder (standard grind) was purchased from H. Reeve Angel. Lactate dehydrogenase (rabbit muscle, crystalline), pyruvate kinase (rabbit muscle, crystalline), NADH, ADP, and trisodium phosphoenolpyruvate were obtained from Sigma. Mercaptoethanol was a product of Eastman Organic Chemicals. ATP was obtained from P-L Biochemicals. 5,5-Dithiobis-(2-nitrobenzoic acid) was obtained from Aldrich. Purified erythrocytic nucleoside diphosphokinase (specific activity, 60 μM units per mg of protein) was a gift from Dr. A. S. Zygmuntowicz. All reagents were of the highest grade of purity commercially available. All commercial samples of inosine tested were found to contain hypoxanthine contamination of 2% or greater. Therefore inosine was recrystallized from water at least three times, and the contamination was reduced to 0.5% or less. Freshly prepared inosine solutions were used whenever needed. Calcium phosphate gel was prepared according to the method of Tsuboi and Hudson (8). Human blood was kindly supplied by Dr. Jacob Dyckman, Director of the Blood Bank of the Miriam Hospital in Providence.

Methods

Spectrophotometric Assay of Phosphorolytic Activities—The assay utilized for most studies was the coupled xanthine oxidase method of Kalckar (11), which is based on the measurement of the increase in absorbance at 293 mμ due to the formation of uric acid. A Beckman DU or a Zeiss spectrophotometer equipped with a Gilford optical density converter and recorder was used.

The reaction mixture for the standard assay contained the following components: inosine, 0.5 μmole; potassium phosphate buffer, pH 7.5, 50 μmoles; xanthine oxidase, 0.02 μM unit; and an appropriate amount of purine nucleoside phosphorylase, in a cuvette having a 1-cm light path, in a final volume of 1 ml, at room temperature. The mixture of all components, except purine nucleoside phosphorylase, was preincubated for about 1 min to remove any trace of hypoxanthine or xanthine introduced as a contaminant in the inosine.

Spectrophotometric Assay of Ribosyl Transfer Reaction—The reaction mixture contained 0.2 μmole of guanine as the recipient of ribose, and 2 μmoles of inosine as the donor of ribose, 0.05 μM unit of xanthine oxidase, and 50 μmoles of Tris-acetate buffer, pH 7.5, in a final volume of 1 ml. Various measures were taken to avoid contamination by inorganic orthophosphate, as noted below. After a preincubation period of 5 min, the reaction was started by the addition of purine nucleoside phosphorylase.

Unit of Purine Nucleoside Phosphorylase—One unit of purine nucleoside phosphorylase is defined as that amount of enzyme which gives an optical density increase of 12.5 per min (1 μmole of uric acid formation) at 293 mμ under standard assay conditions. Similarly, 1 unit of ribosyl transfer activity is defined as that amount of enzyme which catalyzes the formation of 1 μmole of uric acid per min under ribosyl transfer assay conditions.

Measurement of Protein Concentration—Protein concentrations were measured by the biuret procedure (12) or by the method of Warburg and Christian (13), in which the absorbance at 280 mμ was multiplied by a factor of 1.1 to obtain the protein concentration in milligrams per ml.

RESULTS

Purification of Purine Nucleoside Phosphorylase from Human Erythrocytes

Human blood (usually stored for 4 to 6 weeks in a blood bank) was used, and all steps were carried out at 3–5°, unless otherwise stated. The first few steps of purification were adapted from the earlier methods of Tsuboi and Hudson (8) and of Huennekens, Nurk, and Gabrio (9).

Step 1: Preparation of Hemolysate—The plasma was removed by siphoning after the cells had settled. The cells were washed three times by resuspension in 3 volumes of 0.9% sodium chloride solution, followed by centrifugation at 3000 $\times$ *g* for 15 min and removal of the supernatant fluid and the buffy coat. Finally, the packed cells were hemolyzed by the addition of an equal volume of cold distilled water, followed by freezing at −20°. These hemolysates could be stored for lengthy periods and thawed immediately prior to the succeeding step without loss of activity.

Step 2: Calcium Phosphate Gel Fractionation—A 200-ml aliquot of the hemolysate from Step 1 was thawed and diluted with cold, distilled water to a final volume of 1000 ml. Calcium phosphate gel in an amount of 0.05 mg (dry weight) per mg of protein was added, and the mixture was stirred occasionally. After about 30 min, the gel was collected by centrifugation at 2000 $\times$ *g* for 5 min, and washed with cold distilled water by resuspension and centrifugation. The washing was repeated (usually six times) until the supernatant fluid was almost colorless. After the final washing, the enzyme was eluted twice with 100-ml volumes of 0.1 M potassium phosphate buffer, pH 7.5. The enzymatic activity of the hemolysate was recovered almost quantitatively in the eluate.

Step 3: Ammonium Sulfate Fractionation—The enzyme solution from the previous step was brought to 40% saturation by adding solid ammonium sulfate (23.11 g/100 ml of eluate). After the solution had stood for 30 min or longer, the precipitate was removed by centrifugation at 9000 $\times$ *g* for 30 min. Sufficient solid ammonium sulfate (12.31 g/100 ml of supernatant fluid) was added to the supernatant fluid to bring the concentration to 60% saturation. An hour later, the precipitate was collected by centrifugation at 9000 $\times$ *g* for 40 min and redissolved in about 15 ml of 0.1 M Tris-acetate buffer, pH 7.5.

The enzyme solution from the ammonium sulfate fractionation was dialyzed overnight against several changes of 4000 ml of 0.03 M Tris-acetate buffer, pH 7.5. The dialysis casing used in this and other steps was pretreated with hot EDTA solution, as suggested by Tsuboi and Hudson (8).

Step 4: DEAE-cellulose Column Chromatography—The dialyzed enzyme solution was loaded on a DEAE-cellulose column (acetate form, 2.5 $\times$ 12 cm) which had been equilibrated with 0.03 M Tris-acetate buffer, pH 7.5. The column was then washed with 50 ml of the same buffer. During this washing, most of the pink proteins were removed. Then a linear gradient was started with Tris-acetate buffer (pH 7.5) varying from 0.03 to 0.34 M in a total volume of 600 ml. Ten-milliliter fractions were collected. About 95% of the enzyme emerged between tubes 10 and 31, corresponding to concentrations of Tris-acetate of about 0.06 to 0.2 M.

Step 5: Calcium Phosphate Gel-cellulose Column Chromatography—The enzyme solution from the previous step was loaded directly on a calcium phosphate gel-cellulose column (2.5 $\times$ 25

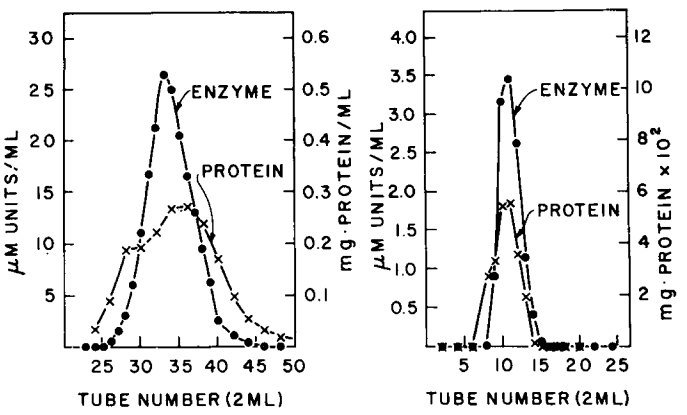


FIG. 1. Purification of human erythrocytic purine nucleoside phosphorylase by Sephadex G-100 column filtration. *Left*, the elution profile of Step 6 described in the text (column size, 1.9 × 50 cm); *right*, the elution profile of the third gel filtration (column size, 1.0 × 55 cm).

cm) which had been equilibrated with 0.1 M Tris-acetate, pH 7.5. The ratio of the cellulose to the gel was 30:1 (dry weight). The elution was performed with a linear gradient of potassium phosphate buffer, pH 7.5, from 0 to 0.1 M in 0.1 M Tris-acetate buffer, pH 7.5, in a total volume of 500 ml. Ten-milliliter fractions were collected. The colored proteins appeared before tube 20, and the enzyme peak emerged between tubes 23 and 32, corresponding to a phosphate concentration of about 0.05 to 0.07 M. The almost colorless fractions containing the enzyme were pooled and concentrated by adding solid ammonium sulfate (40.66 g/100 ml) to a saturation of 65%.

Step 6: Sephadex G-100 Column—The precipitates in the ammonium sulfate suspension were collected by centrifugation at 10,000 × *g* for 40 min, dissolved in a small amount (1 to 2 ml) of 0.1 M Tris-acetate buffer, pH 7.5, and transferred to a Sephadex G-100 column (1.9 × 50 cm) equilibrated with 0.1 M Tris-acetate, pH 7.5. The filtration was carried out with the same buffer, and 2-ml fractions were collected. The enzyme emerged between tubes 30 and 37, as shown in Fig. 1. The pooled enzyme solution could be concentrated for storage by lyophilization or by 80% saturation with ammonium sulfate.

The results of a typical preparation are shown in Table I. If desired, the enzyme may be purified further by repeating the DEAE-cellulose and Sephadex column steps, or the latter step alone. By repeating the Sephadex column step twice more, a preparation with a specific activity of 80 was obtained. Also shown in Fig. 1 is that the enzyme and protein emerged from the third Sephadex column (packed in a 50-ml burette) in single peaks which coincided closely.

pH Optimum

Previous workers have reported human erythrocytic purine nucleoside phosphorylase to have a pH optimum of 6.8 to 7.0 (14) and 7.0 to 8.5 (9) for the phosphorolytic reaction and of 6.0 for the arsenolytic reaction (8).

With the enzyme preparations examined here by means of the standard assay procedure, a broad pH optimum range of about 6.5 to 8.0 was observed. The pH chosen for kinetic and other studies was 7.5. An investigation of the influence of pH on the kinetic parameters with different substrates and for the arsenolytic, phosphorolytic, and ribosyl transfer reaction has been undertaken.

Estimation of Molecular Weight

The estimation of the molecular weight was performed according to the method of Andrews (15) with a Sephadex G-200 column (1.9 × 50 cm), which had been equilibrated with 0.1 M Tris-acetate buffer, pH 8.0, containing 1×10^{-4} M dithiothreitol. A mixture of pyruvate kinase, lactate dehydrogenase, erythrocytic nucleoside diphosphokinase, bovine serum albumin, and purine nucleoside phosphorylase in a 1.5-ml total volume was filtered through the Sephadex G-200 column. The tube numbers of the enzyme peaks and of the highest absorbance at 280 mμ for the albumin peak were plotted against the logarithm of the known molecular weights of the proteins. The molecular weight of the erythrocytic purine nucleoside phosphorylase was estimated as approximately 80,000 (Fig. 2).

Failure to Separate Phosphorolytic and Ribosyl Transfer Activities

Enzyme preparations at various stages of purity (from 3.0 to 61 μM units per mg of protein) were examined for the relative amounts of phosphorolytic and ribosyl transfer activities. Various methods were utilized to remove possible trace contaminants of orthophosphate: filtration through a Sephadex G-100 column; exhaustive dialysis against 2.5 mM Tris-acetate or Tris-maleate, pH 7.5; or filtration through a Dowex 1-chloride anion exchange column. In the last case, all reagent solutions were passed through the Dowex 1-chloride column. Regardless of the method used to remove phosphate contaminant, and regardless of the stage of purity, the ratios of phosphorolytic and ribosyl transfer activities remained in the same order of magnitude (Table II). In these experiments, the reaction mixtures for the measurement of phosphorolytic activity contained 0.2 μmole of guanine in order to compensate for the inhibitory effect of the base both on purine nucleoside phosphorylase and on xanthine oxidase, thus making the comparison more valid.

In an attempt to show evidence for separation of the two activities, a highly purified sample of enzyme (specific activity, 61 μM units per mg) was filtered through a Sephadex G-100 column, and aliquots from the head, middle, and tail of the enzymatic

TABLE I

Purification of erythrocytic purine nucleoside phosphorylase

Step of purification	Total volume	Total activity of enzyme	Specific activity	Purification	Recovery
	ml	*units*	*units/mg*	*-fold*	*%*
Hemolysate (1:10)	1000	880	0.011	1	100
Calcium phosphate gel fractionation	195	780	0.326	30	89
Ammonium sulfate fractionation (40 ~ 60%)	10.5	378	2.170	197	42
DEAE-cellulose (acetate) column	100	340	7.70	700	38
Calcium phosphate gel-cellulose column	67	330	29.0	2660	37
Sephadex G-100 column	17.5	308	61.2	5560	35

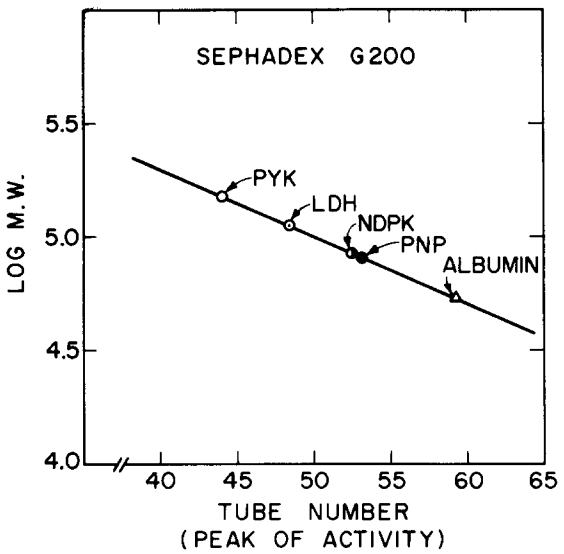


FIG. 2. Estimation of molecular weight of erythrocytic purine nucleoside phosphorylase. The following proteins were mixed in 1.5 ml of 0.1 M Tris-acetate, pH 7.5, containing dithiothreitol, 1×10^{-3} M, and were loaded on to a Sephadex G-200 column: 30 μM units of pyruvate kinase (*PYK*; mol wt 230,000; specific activity, 250), 30 μM units of lactate dehydrogenase (*LDH*; mol wt 120,000; specific activity, 450), 50 μM units of erythrocytic nucleoside diphosphokinase (*NDPK*; mol wt 100,000; specific activity 60), 50 mg of bovine serum albumin (mol wt 67,000), and 30 μM units of erythrocytic purine nucleoside phosphorylase (*PNP*) (specific activity, 40).

TABLE II

Comparison of phosphorolytic and ribosyl transfer activities of human erythrocytic purine nucleoside phosphorylase at various stages of purity

The reaction mixture for the phosphorolytic activity in this experiment contained the same components as those for ribosyl transfer activity, in addition to 50 mM phosphate.

Specific activity	Phosphorolytic activity (A)	Ribosyl transfer activity (B)	B:A	Pretreatment and buffer
units/mg	*units/ml*	*units/ml*		
61.0	1.38	0.005	0.0037	Sephadex, Tris-acetate
	1.48	0.012	0.0081	Sephadex + dithiothreitol, Tris-acetate
30.0	0.96	0.008	0.0082	Dialysis, Tris-maleate
	3.56	0.013	0.0037	Dowex 1, Tris-acetate
25.0	0.42	0.0012	0.0030	Dialysis, Tris-maleate
20.0	0.84	0.0047	0.0056	Sephadex, Tris-maleate
16.0	3.16	0.026	0.0080	Sephadex, Tris-acetate
10.0	2.12	0.005	0.0020	Sephadex, Tris-acetate
4.5	1.20	0.007	0.0058	Dialysis, Tris-maleate
	0.62	0.003	0.0048	Dialysis, Tris-maleate
3.7	1.88	0.014	0.0075	Dialysis, Tris-maleate
3.0	0.54	0.005	0.0093	Dialysis, Tris-acetate
	1.70	0.012	0.0070	Dialysis, Tris-acetate

peak were examined for phosphorolytic and ribosyl transfer activities. The ratio of the two activities remained the same across the peak.

The above experiments indicate that the phosphorolytic and ribosyl transfer activities are catalyzed by the same molecular species. Of course it will be necessary to re-examine the question when crystalline preparations of erythrocytic purine nucleoside phosphorylase become available. A sample of crystalline calf spleen purine nucleoside phosphorylase (specific activity, 30 to 40 μM units per mg) (Boehringer Mannheim) was found to have a ratio of the two activities similar to those observed with the human erythrocytic enzyme preparations studied here.

Stability of Enzyme

As is suggested by the high yield during purification, erythrocytic nucleoside phosphorylase is a reasonably stable enzyme. The enzyme may be stored without loss of activity as a lyophilized powder, or as a suspension in 80% saturated ammonium sulfate for several months. It also has been stored as a dilute solution in 0.1 M Tris-acetate buffer, pH 7.5, for 1 month at 4° with less than 15% loss of activity. As may be seen in Fig. 3, the activity decays within a few days if the enzyme is stored in 2.5 mM Tris-acetate, pH 7.5, in the absence of a thiol-protective reagent. Treatment of the enzyme with mercaptoethanol (Fig. 3) protects it from inactivation by incubation at low ionic strength, whereas incubation with dithiothreitol on several occasions has resulted first in an increase in activity, followed by a marked inactivation.

The data presented in Table III show that reagents which react with thiol groups, such as *p*-chlormercuribenzoate and 5,5′-dithiobis(2-nitrobenzoic acid), inactivate the enzyme. Enzyme inactivated by either of these reagents may be partially reactivated by treatment with 7 mM dithiothreitol. The unexplained qualitative difference in the effect of the two thiols, mercaptoethanol and dithiothreitol, on the enzyme, and the evidence of an important involvement of thiol groups in enzymic activity, are deserving of further detailed study.

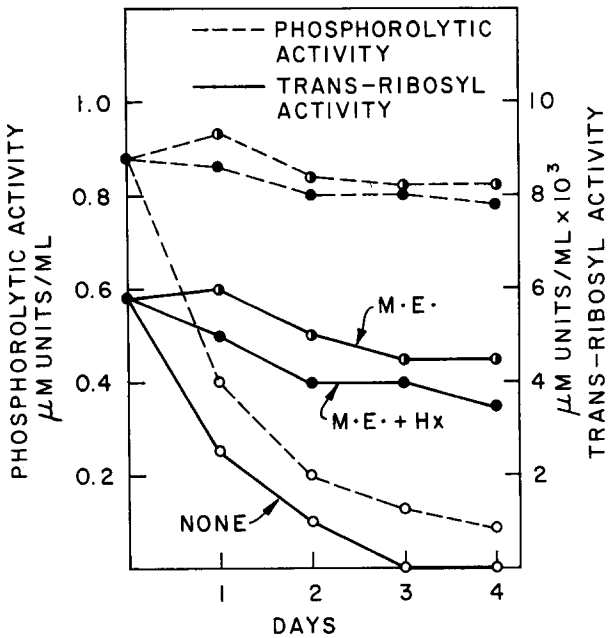


FIG. 3. Protection of phosphorolytic and ribosyl transfer (or base exchange) activities from inactivation in the presence of mercaptoethanol. The enzyme from a Sephadex G-100 column (specific activity, 70) was dialyzed overnight against 2.5 mM Tris-acetate buffer, pH 7.5, and then allowed to stand at 4° in the presence of 10 mM mercaptoethanol (*M.E.*) with or without 0.8 mM hypoxanthine (*Hx*).

TABLE III

Effect of thiol reagents on purine nucleoside phosphorylase

Purine nucleoside phosphorylase (specific activity, 30; 0.6 μM unit per ml) in 0.1 M Tris-acetate buffer, pH 7.5, was incubated at room temperature with 5,5′-dithiobis(2-nitrobenzoic acid) or *p*-chlormercuribenzoate at a final concentration of 2 mM. At 60 min of the incubation, dithiothreitol (final concentration, 7 mM) was added to all samples.

Reagent	Activity remaining after				
	1 min	30 min	70 min	100 min	15 hrs
	%	%	%	%	%
None	100	94	98	88	88
5,5′-Dithiobis(2-nitrobenzoic acid)	75	8	14	32	33
p-Chloromercuribenzoate	20	17	75	72	60

Failure To Demonstrate Interconversion of Phosphorolytic and Ribosyl Transfer Activities

Recently Pinto and Touster (6) described an apparent conversion of phosphorolytic to ribosyl transfer activity when purine nucleoside phosphorylase from Ehrlich ascites cells was allowed to stand at 4° for 1 day or more in the absence of mercaptoethanol, or in the presence of both mercaptoethanol and hypoxanthine. They also showed two peaks of ribosyl transfer activity with an intermediate peak of phosphorylase activity by gradient density centrifugation.

As may be seen in Fig. 3, when an experiment essentially similar to that of Pinto and Touster was performed with purified erythrocytic purine nucleoside phosphorylase, a marked decline in activity occurred when the enzyme was incubated at low ionic strength in the absence of mercaptoethanol. However, the phosphorolytic and ribosyl transfer activities decreased proportionally. Furthermore, in contrast to the findings of Pinto and Touster, when the erythrocytic enzyme was incubated with mercaptoethanol and hypoxanthine for 4 days, only a slight decrease in activity was observed. In a number of studies with partially inactivated enzyme, there was no indication of the separation of phosphorolytic and ribosyl transfer activities by means of filtration through Sephadex columns.

Effect of Alternative Competing Substrates

The effect of alternative competing substrates on the reaction velocity was studied in two different situations. In one case the formation of a product derived from only one substrate was determined, while in the other case the combined rate of formation of the product from both alternative substrates was determined.

The ribo- and deoxyribonucleoside of guanine were found to behave as competitive inhibitors when either the ribonucleoside or deoxyribonucleoside of hypoxanthine was the variable substrate at a constant level of phosphate, and the rate of hypoxanthine formation was measured, as shown in Figs. 4 through 7. These findings, although not conclusive, are consistent with the hypothesis that a single enzyme acts upon all four nucleosides tested: inosine, deoxyinosine, guanosine, and deoxyguanosine.

The determination of the combined rates of formation of hypoxanthine from both inosine and deoxyinosine comprises the second situation. When the concentration of either inosine or deoxyinosine was varied over a wide range at a constant level of phosphate, the Lineweaver-Burk plots for both substrates yielded straight lines at low concentrations, but concave downward curves at high concentrations of the substrate, as shown in Fig. 8. Several possibilities may be considered to explain these nonlinear plots. Among these is the possibility of the occurrence of alternate reaction sequences, *i.e.* a "random Bi Bi mechanism" according to Cleland's nomenclature (16). However, this mechanism, in which the time sequence both of the binding of substrates and of the dissociation of products is random, seems very unlikely in view of the product inhibition data presented in the

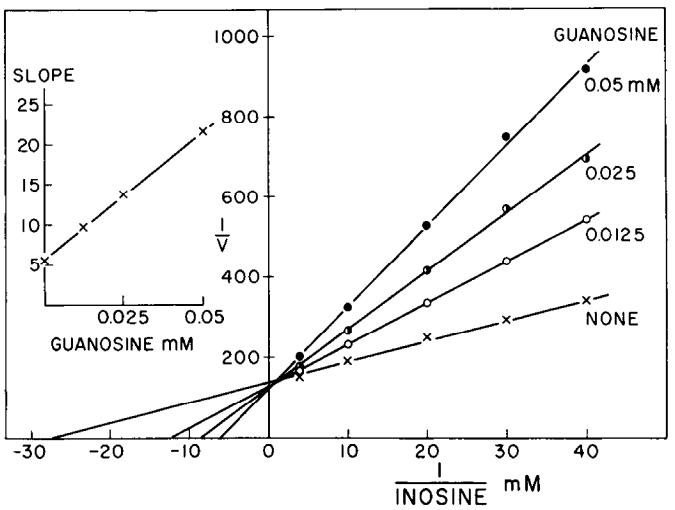


FIG. 4. Inhibition by the alternative substrate, guanosine, when inosine was the variable substrate. Plot of reciprocal of the initial velocity (micromoles of uric acid per min per ml) with respect to reciprocal of inosine concentration (millimolar). One milliliter of the reaction mixture contained 50 μmoles of potassium phosphate buffer, pH 7.5, and 0.02 μM unit of xanthine oxidase. Inosine and guanosine were added in the concentrations indicated. The inhibition constant, K_i, of guanosine as the competitor of inosine, estimated by the replot shown in the *inset*, was 1.5×10^{-5} M.

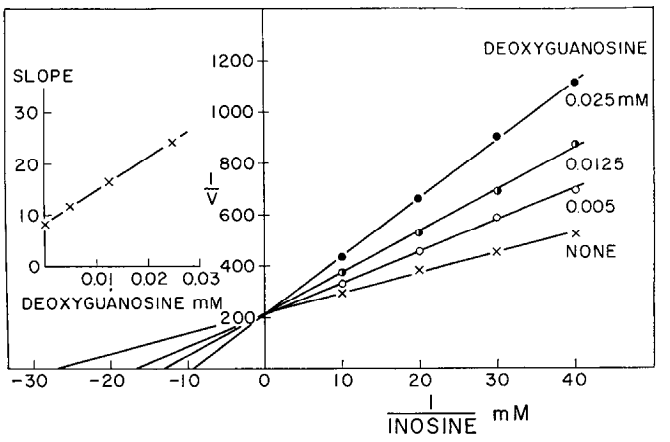


FIG. 5. Inhibition by the alternative substrate, deoxyguanosine, when inosine was the variable substrate. Plot of reciprocal of the initial velocity (micromoles of uric acid per min per ml) with respect to reciprocal of inosine concentration (millimolar). One milliliter of the reaction mixture contained 50 μmoles of potassium phosphate buffer, pH 7.5, and 0.02 μM unit of xanthine oxidase. Inosine and deoxyguanosine were added in the concentrations indicated. The inhibition constant, K_i, of deoxyguanosine as the competitor of inosine, estimated by the replot shown in the *inset*, was 1.3×10^{-5} M.

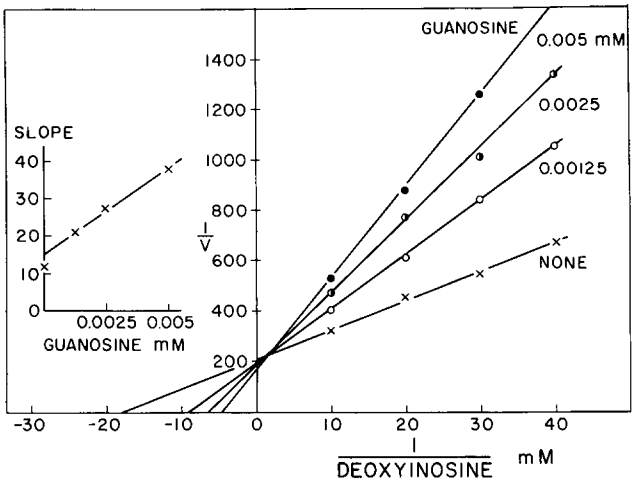


FIG. 6. Inhibition by the alternative substrate, guanosine, when deoxyinosine was the variable substrate. Plot of reciprocal of the initial velocity (micromoles of uric acid per min per ml) with respect to reciprocal of deoxyinosine concentration (millimolar). One milliliter of the reaction mixture contained 50 μmoles of potassium phosphate buffer, pH 7.5, and 0.02 μM unit of xanthine oxidase. Deoxyinosine and guanosine were added in the concentrations indicated. The inhibition constant, K_i, of guanosine as the competitor of deoxyinosine, estimated by the replot shown in the *inset*, was 3.7×10^{-6} M.

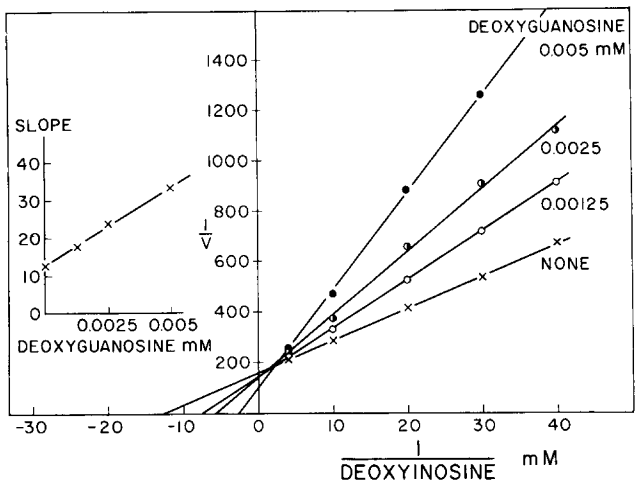
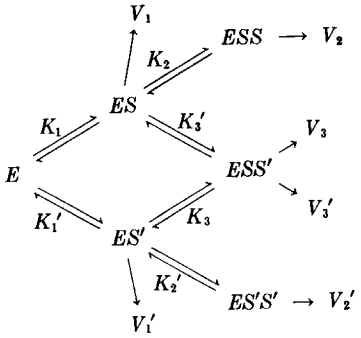


FIG. 7. Inhibition by the alternative substrate, deoxyguanosine, when deoxyinosine was the variable substrate. Plot of reciprocal of the initial velocity (micromoles of uric acid per min per ml) with respect to reciprocal of deoxyinosine concentration (millimolar). One milliliter of the reaction mixture contained 50 μmoles of potassium phosphate buffer, pH 7.5, and 0.02 μM unit of xanthine oxidase. Deoxyinosine and deoxyguanosine were added in the concentrations indicated. The inhibition constant, K_i, of deoxyguanosine as the competitor of deoxyinosine, estimated by the replot shown in the *inset*, was 3.8×10^{-6} M.

accompanying paper (17). It is also possible that there is more than one enzyme, or that some sort of allosteric effect occurs. Also shown in Fig. 8 are the experimental points obtained by varying the inosine concentration at a constant level of phosphate in the presence of a constant amount of deoxyinosine, which gave similar nonlinear double reciprocal plots.

Of the several possible models that might explain the data in Fig. 8, the following mechanism has been carefully studied. The model postulates that the enzyme has two catalytic sites which interact with each other, and the reaction pathway may be described as shown in Equation 2. The rate equation derived on



(2)

the basis of equilibrium assumptions, by a method to be published elsewhere (18), is

$$v = \frac{\frac{V_1(S)}{K_1} + \frac{V_2(S)^2}{K_1K_2} + \frac{(V_3 + V_3)(S)(S')}{K_1K_3'} + \frac{V_1'(S')}{K_1'} + \frac{V_2'(S')^2}{K_1'K_2'}}{1 + \frac{(S)}{K_1}\left(1 + \frac{(S)}{K_2}\right) + \frac{(S)(S')}{K_1K_3'} + \frac{S'}{K_1'}\left(1 + \frac{S'}{K_2'}\right)} \quad (3)$$

The parameters estimated by the least square fit of the data to Equation 3 are listed in Table IV, and the theoretical lines based on those parameters are also shown in Fig. 8 (*Lines 1, 2,* and *4*).

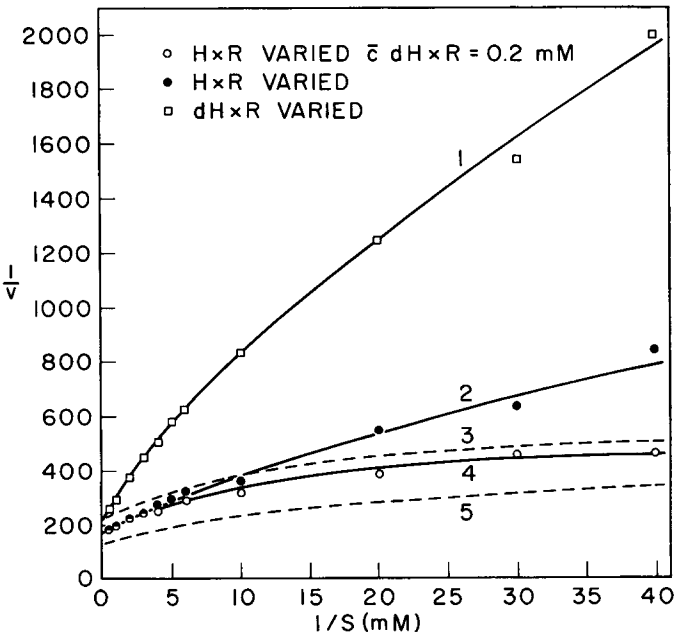


FIG. 8. The velocity of purine nucleoside phosphorylase reaction as a function of concentrations of the substrates inosine, deoxyinosine, or both. The concentration of phosphate was held constant at 50 mM. □, rate of formation of hypoxanthine with deoxyinosine as the variable substrate; ●, rate with inosine as the variable substrate; ○, combined rates with various concentrations of inosine in the presence of a constant amount (0.2 mM) of deoxyinosine. *Lines 1, 2,* and *4* were drawn by calculations based on Equation 3 and the parameters estimated by the least squares method (Table IV). *Line 3* represents the univalent nonspecific enzyme model, and it was drawn on the basis of Equation 4 and approximate kinetic parameters of $V = 0.0045$ μmole per min, $V' = 0.0025$ μmole per min, $K = 0.07$ mM, and $K' = 0.1$ mM. *Line 5* was calculated by adding a constant 0.0017 μmole per min (the velocity with 0.2 mM deoxyinosine) to the velocities observed with inosine alone.

If the enzyme has only one catalytic center which reacts with both inosine and deoxyinosine, or if the enzyme has multiple catalytic sites which do not interact with each other, the combined velocities of the reaction with inosine and deoxyinosine are given by

$$v = \frac{\frac{VS}{K} + \frac{V'S'}{K'}}{1 + \frac{S}{K} + \frac{S'}{K'}} \quad (4)$$

where v is the combined rate, V, K, and S are the maximal velocity, Michaelis constant, and the concentration of inosine, respectively, and V', K', and S' are the corresponding symbols for deoxyinosine. *Line 3* in Fig. 8 was drawn by calculation on the basis of Equation 4, with the kinetic parameters estimated by extrapolation of the linear portions of *Lines 1* and *2*. As may be seen in the figure, *Line 3*, representing the univalent nonspecific enzyme model, agrees very poorly with the experimentally observed combined velocities.

It is quite conceivable that, despite the purification procedure, the enzyme preparation might consist of a mixture of several enzymes, each specific for a nucleoside. *Line 5* in Fig. 8 was drawn on the basis of such a hypothesis: that there are two different enzymes, one specific for inosine and the other for deoxyinosine.

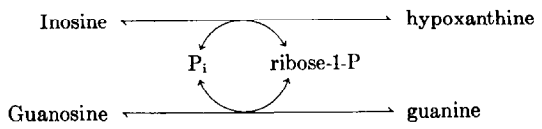
$$v = \frac{VS}{K + S} + \text{constant} \quad (5)$$

where the constant is the velocity with 0.2 mM deoxyinosine alone. It is clear that the prediction of a two-enzyme model does not agree with the observed data.

DISCUSSION

By the relatively simple procedure described in this paper, purine nucleoside phosphorylase may be isolated from human erythrocytes in excellent yield, and in a high degree of purity. Although rigorous criteria have not yet been applied, it appears that preparations with specific activities of 80 μM units per mg are close to homogeneity. It is of interest that in an earlier study a preparation of enzyme from human erythrocytes with a specific activity of 21 μM units per mg appeared homogeneous by electrophoresis and ultracentrifugation (8). Moreover, a crystalline preparation of purine nucleoside phosphorylase from calf spleen with a specific activity of 30 to 40 units per mg is available commercially.

The question of the occurrence and significance of the ribosyl transfer reaction catalyzed by purine nucleoside phosphorylase is an intriguing one. From the studies reported above, it appears that the phosphorylase and ribosyl transfer reactions are catalyzed by a single enzyme. However, it should be noted that a small amount of phosphate could catalyze an apparent ribosyl transfer reaction by the following mechanism, in which orthophosphate would be continually regenerated in a cyclic manner.



Several considerations lead us to believe that an inadvertent contamination by phosphate could not have occurred in these experiments. First, in the preincubation procedures used, a phosphate contamination of as little as 0.5 nmole per ml would have been detected. The phosphorolytic to ribosyl transfer reaction ratio was not significantly affected by treatment of all solutions with Dowex 1-chloride or by prolonged dialysis of the enzyme preparations. In an experiment described in detail in the accompanying paper (17), a complete reaction mixture containing purine nucleoside phosphorylase and ^{32}P-orthophosphate was filtered through a Sephadex G-100 column. No orthophosphate traveled with the enzyme, which indicates that the Sephadex filtration procedure used in purification cleanly removes orthophosphate from the enzyme protein.

The above experiments describe the inability of this laboratory to demonstrate the conversion of phosphorolytic to ribosyl transfer activity reported by Pinto and Touster for purine nucleoside phosphorylase isolated from Ehrlich ascites cells. The reason for this discrepancy is not apparent; however, it seems possible that the difference in enzyme source or a difference in the techniques used by the two laboratories is responsible.

The study of the effect of alternative competing substrates, particularly the study of rate as a function of inosine concentration in the presence or absence of deoxyinosine, presented in Fig. 8, is worthy of attention. A comparison of the experimental points (the *open circles* along *Line 4*) with *Lines 3* and *5* effectively rules out two possibilities. The first is that there are two enzymes, one highly specific for inosine and the other highly specific for deoxyinosine, neither of which is affected by the presence of the substrate for the other (*Line 5*). The second possibility which may be ruled out is that there is a single enzyme reactive with both inosine and deoxyinosine, which compete with each other for a single site on the enzyme (or for multiple catalytic sites which do not interact with each other). Equation 3, which is based on two catalytic sites with two competing alternative substrates, seems to be in harmony with the experimental data. It must be pointed out, however, that the mere fact that the data may fit Equation 3 does not necessarily indicate that the enzyme has two interacting catalytic centers. It may have more than two catalytic centers. Another important possibility which cannot be ruled out at the present time is that we are dealing with isozymes which are so similar in their chemical and physical properties as to have resisted separation by the techniques used to date. If such isozymes occur, they must differ

TABLE IV

Kinetic parameters estimated on assumption that purine nucleoside phosphorylase has two interacting catalytic centers

The calculations were carried out by fitting the experimental points in Fig. 8 to Equation 3 by the least squares method. The parameters for deoxyinosine as the substrate may be considered as those with primes in Equation 3. K values are expressed in millimoles per liter; V values, in nanomoles per min per ml.

Parameters	Variable substrate	
	Inosine	Deoxyinosine
K_1	0.018	0.022
K_2	0.23	0.52
K_3	0.14	0.17
V_1	1.7	0.79
V_2	5.8	4.7
$V_3 + V'_3$	5.5	

significantly in their kinetic parameters. Additional examination of this question in the course of further attempts at purification with the application of different techniques, such as electrophoresis, is indicated.

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